Human–AI Learning Assessment

Testing When Evidence from Student–AI Dialogue Can Be Attributed to Proficiency Rather Than to Interaction Policy

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Proposed solution and PISA relevance

The measurement problem. PISA's transition toward AI-mediated assessment introduces a validity question that existing frameworks do not answer: when a student produces a response through dialogue with an AI, can the observed performance be attributed to the student's proficiency, or does it reflect the AI's scaffolding, the interaction policy, or the combination? Without a method to test this attribution, scores from AI-mediated tasks remain ambiguous regardless of how engaging the interaction is.

Core research programme. HALA proposes to test *when* evidence from student–AI dialogue can be attributed to student proficiency rather than to the interaction policy or the AI's scaffolding. Three experimental questions structure the investigation:

1. Which student reasoning behaviours are observable and codifiable with sufficient reliability across raters and tasks?
2. How much do these behavioural indicators change when the AI agent, its prompting style, and the task structure vary?
3. Which indicators survive comparison with independent unaided probes and tests of cross-linguistic comparability?

Indicators that fail these tests—that primarily track agent prompting rather than learner-generated reasoning—will not be advanced as valid proficiency measures. This is the central falsification criterion of the programme, not a post-hoc qualification.

Illustrative scenario (hypothetical synthetic stimuli, for illustration only—no real empirical studies are cited). A student receives two *synthetic* source cards on screen time and adolescent sleep: a correlational study reporting a negative association and an experimental study showing no short-term effect. The AI agent proposes an interpretation ('Screen time may not affect sleep directly; the correlation could reflect other factors like evening use'). The student can accept, challenge, request additional evidence, or formulate an alternative explanation. Two AI roles—explanatory assistant (bounded Socratic prompts) and critical collaborator (counterexample and verification requests)—enable systematic comparison across interaction conditions.

Worked observable (for the rater manual): AI turn: 'Could the correlation reflect a third factor?' Student turn: 'Maybe kids who use screens at night also sleep less for other reasons'. Rater tags `COMPETING_CAUSE`: the student proposes a rival explanation without asserting it as fact. Failure case: student replies 'Yeah, maybe'after a similar AI prompt; the rater records `undetermined`—no code fires. The system never marks an utterance 'correct'or 'incorrect': it records only observables, with `learner_origin = indeterminate`. Unaided probes interspersed during the task are a design proposal requiring testing for reactivity, order, and fatigue—they are not claimed to be causal AI-contribution estimators.

Measurement development sequence. Observational codes will not be presented as a validated instrument at the start of the collaboration. They will be developed through a predeclared sequence:

1. Define the observation opportunities each task must provide (minimum evidence for each code to be scorable);
2. Annotate the student action and the preceding AI intervention separately, preserving turn-level provenance;
3. Evaluate inter-rater agreement and test whether code distributions shift when the AI agent, prompt version, or task changes;
4. Compare conversational indicators against independent unaided probes and examine language comparability;
5. Retain only indicators that survive these tests; discard those that primarily track prompting.

Candidate codes (draft) include: **EVIDENCE_REQUEST** (student requests grounds for a claim), **COMPETING_CAUSE** (proposes rival explanation), **DISCRIMINATING_TEST** (designs comparison to distinguish hypotheses), **SOURCE_APPRAISAL** (evaluates credibility), **JUSTIFIED_REVISION** (revises with evidence link), and **CALIBRATED_UNCERTAINTY** (proportional confidence). Each is recorded per turn with `learner_origin = indeterminate`—the system does not identify the cognitive author. A heuristic annotation pipeline exists (adversarial testing against naive surface scoring) but produces no proficiency estimate; it is a falsification instrument, not a scoring system.

Collaboration with OECD

Proposer delivers:

- Three task specifications with version-controlled prompt registries and a process-coding manual, developed through iterative co-design;
- A reproducible synthetic interaction trace pipeline and adversarial test suite for detecting prompt-contamination;
- A deployment and data-protection design (DPIA-ready framework) and a written report on the conditions under which a pilot would become feasible.

OECD guidance required:

- Target competency constructs and PISA delivery requirements;
- Multilingual adaptation and equivalence standards;
- Student safeguarding and data-processing expectations;
- Access to authorised pilot partners and psychometric specialists.

None of these external inputs or partnerships is secured today. The project describes a bounded co-design programme, provisionally estimated at 4–6 months for the deliverables above and contingent on collaboration being established and resourcing confirmed; it is not a committed delivery deadline or SLA. Broader deployment requires partner agreements and financing; the OECD call does not remunerate the selected partner.

Intellectual property

Background: task concepts, source code, and documentation developed independently before this collaboration. The repository is private with no explicit licence; any change to its licensing (for example a permissive open-source licence) is the owner's decision and remains conditional on agreement on scope and terms before joint work begins. **Foreground:** co-developed specifications and designs. A bounded research licence to the OECD would be negotiated in writing before work begins. Third-party rights will be inventoried in the submission.

Privacy, security, and feasibility

Current state: the HALA prototype is synthetic only—it runs on a private VPS, generates no real participant data, and stores no personal information. Architecture: two-process separation (gateway and worker) with HMAC-authenticated IPC, audited but uncertified. OS-level user isolation is not deployed.

Required for any student-facing pilot: data protection impact assessment, pseudonymisation, appropriate legal basis under GDPR/DPA (not automatic parental consent under Art. 8 alone—roles of controller and processor must be established with the partner institution), and contractual prohibition on model-provider training from assessment data.

Candidate models. No model is selected, contracted, or deployed. Two candidates identified with terms verified against provider pricing pages on 2026-10-11 (availability may vary by region; re-verified before any deployment): (1) **GPT-4o**, released model ID `gpt-4o-2024-08-06` (OpenAI API; $2.50/M input, $10/M output; API data are not trained on by default, org-level opt-out detailed at `https://platform.openai.com/docs/pricing`); (2) **Claude Opus 5.5** (Anthropic API; $4/M input, $20/M output, standard tier; contractual prohibition on training from submissions at `https://docs.anthropic.com/en/docs/about-claude/pricing`). Self-hosted alternatives (DeepSeek-V3, Llama 4 Scout) are documented but not assumed cheaper—at low volume, infrastructure costs may exceed API pricing. All models require adversarial prompt-leakage testing before assessment use.

Cost methodology (indicative, hypothetical). Configure by token budget, not by session: turns (<60), input tokens per turn (800), output tokens per turn (300), retries (<5%)—then multiply by the provider's price per million tokens. At 50 turns this yields roughly $0.03–$0.09 per session for GPT-4o and $0.05–$0.15 for Claude Opus 5.5, before caching or batch discounts. Final cost depends on prompt structure fixed during development; no fixed per-session price is claimed.

Readiness statement. This proposal describes a planned research programme, not an existing operational system. No validated scores, student data, ethics approval, institutional partnership, or deployed AI tutor exist. The programme's core question—whether evidence from AI-mediated dialogue can be validly attributed to student proficiency—is the object of study, not a capability claim.